



GTC4Lusso USA - Technical specifications - Latest Update: 2021-01-26

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F3.01 Technical specifications

Direct injection

Overview

Direct injection is a fuel system in which fuel is injected via one or more precisely oriented jets directly into the combustion chamber. The injectors are located within the combustion chamber of each cylinder, so that fuel is injected directly into the individual combustion chamber.

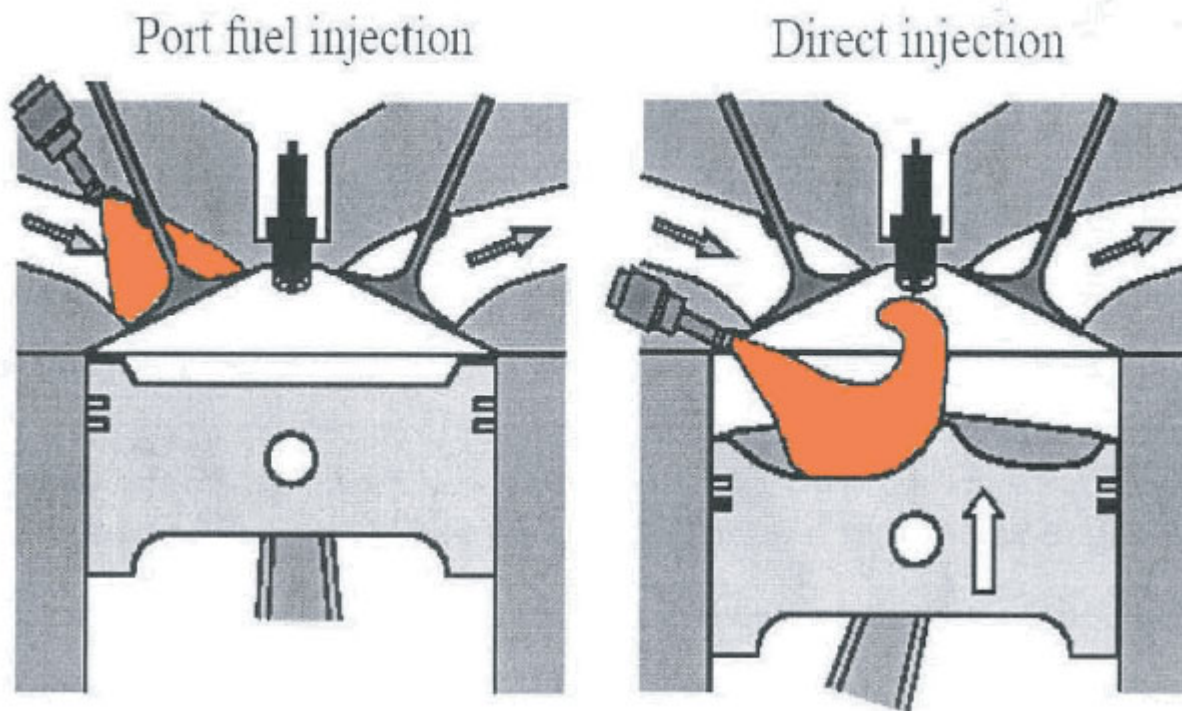
In conventional indirect injection systems, fuel is injected into the stream of aspirated air, before it enters the cylinder. Single point injection systems have a single injector - even for a multi-cylinder engine - located upstream of the point where the intake ducts branch out towards each cylinder. **Multi point** or **PFI (Port Fuel Injection)** systems, on the other hand, have multiple injectors - one per cylinder - located in the intake manifold, ahead of the intake valve. As a result, fuel is mixed differently - in PFI systems, mixing occurs in the intake duct, where part of the fuel evaporates, forming a fine mixture with the air, and part of the fuel does not evaporate, depositing instead on the inner wall of the duct creating a thin liquid layer (a phenomenon known as wall wetting), whereas in the case of direct injection, as only air flows through the ducts, mixing occurs directly within the combustion chamber.

Moreover, direct injection enables a **stratified charge**, resulting in a nonhomogeneous distribution of the fuel in the combustion chamber; in particular, this creates a zone with a stoichiometric or slightly rich mixture zone around the spark plug electrodes. This ensures combustion even with very lean mixtures: the **air/fuel ratios** - usually indicated with the symbol α - is in these cases extremely high, around **40:1**.

Stratification is controlled by a number of factors enabling the confinement of the stoichiometric zone, directing the spray and impeding excessive diffusion throughout the chamber:

- Combustion chamber geometry
- Conformation of the airflow field
- Injector geometry
- Injection type

Stratification allows control over the engine charge by metering the quantity of fuel injected alone - as occurs in diesel engines (quantity regulation) - eliminating the throttle valve and, as a consequence, reducing pumping losses. Furthermore, in low load conditions, the mixture core in which combustion propagates is surrounded by air alone. This insulates it from the surrounding environment, reducing thermal losses.



The following is a summary of the advantages and disadvantages of direct fuel injection.

- Advantages -

- Low fuel consumption and reduced emissions
- High compression ratios
- Improved engine flexibility and response
- High power from smaller displacement engines (if used in conjunction with a turbocharger)

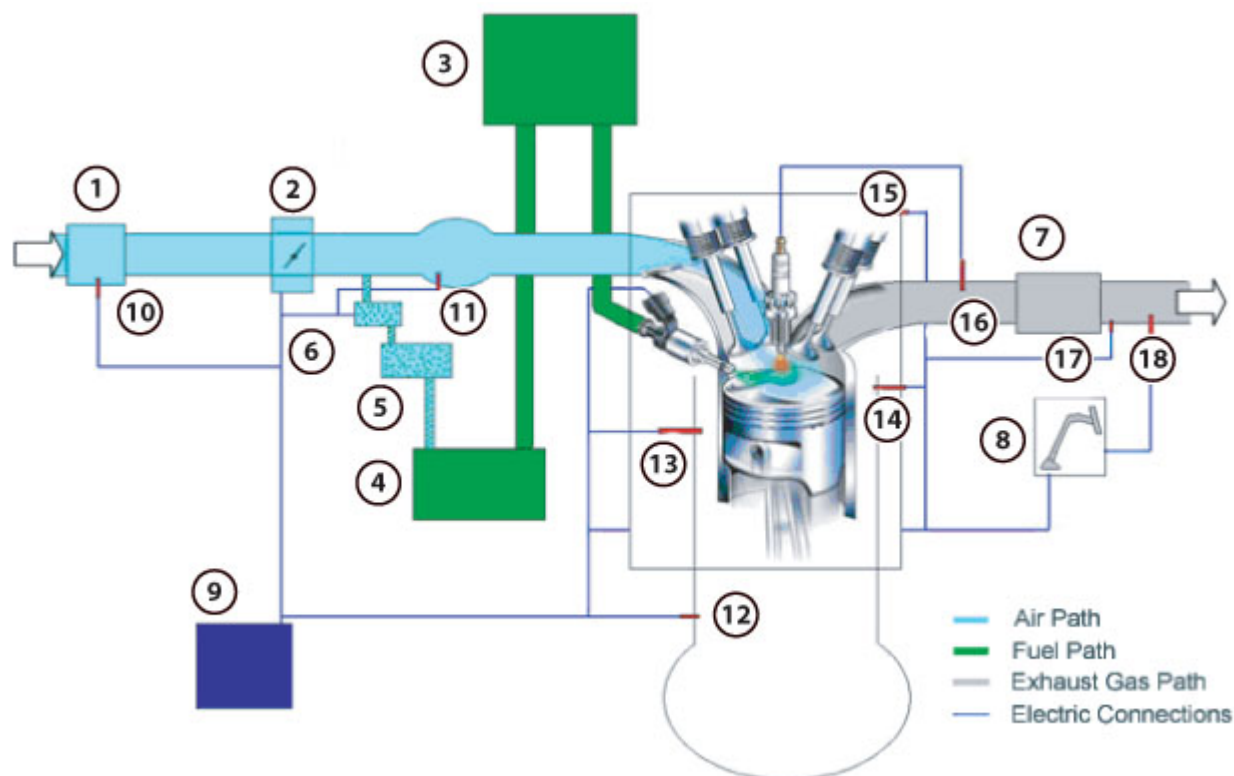
- Disadvantages -

- More complex production process
- Increased costs
- More difficult, more costly repairs

Functional layouts of the Ferrari direct injection system

The engine developed for the Ferrari California unequivocally confirms the validity of direct injection technology in high performance applications.

The implementation of direct injection has necessitated a number of significant changes to the engine and to the head in particular, with the adoption of high pressure pump, different injectors and, obviously, changes to the electronic engine management. The primary components of the direct injection system are shown in the following diagram.

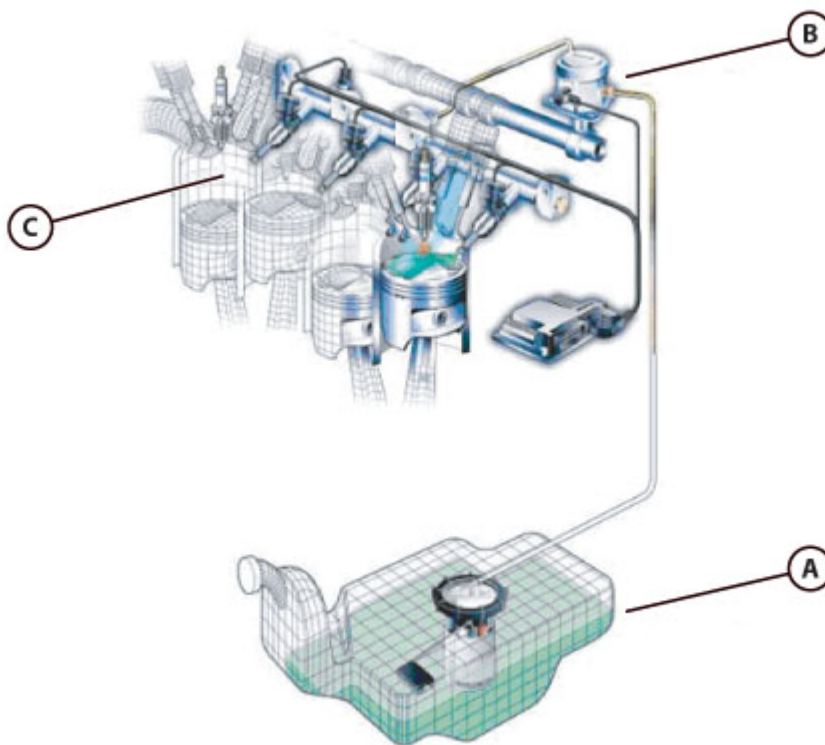


- 1 - Air flow meter
- 2 - Motorised throttle valve
- 3 - High-pressure pump
- 4 - Fuel tank
- 5 - Active carbon filter
- 6 - Bleed valve
- 7 - Trivalent catalytic converter
- 8 - Accelerator pedal
- 9 - ECU
- 10 - Air flow sensor
- 11 - Intake manifold sensor
- 12 - RPM sensor
- 13 - Knock sensor
- 14 - Temperature sensor
- 15 - Timing sensor
- 16 - Proportional oxygen sensor
- 17 - Exhaust gas temperature sensor
- 18 - Potentiometric oxygen sensor

The different subsystems forming the system can be seen in this diagram: intake, fuel supply, exhaust and electrical circuit. On the intake side, once past the air flow meter, the air flows through the throttle valve (unlike conventional systems, this is no longer wet by fuel). Depending on the instantaneous accelerator pedal angle and in relation to the engine mapping, the ECU sends the necessary signal to the injector to deliver the exact amount required. On the exhaust side, note the two oxygen sensors, one ahead of and one after the catalytic converter. The fuel system includes a high pressure pump separating the circuit into two parts: a high pressure section and a low pressure section.

The lubrication and cooling circuits are also of fundamental importance for the engine. The oil circuit features wet sump lubrication, which reduces maintenance costs and times compared with the previous dry sump configuration, and also reduces noise emissions; the oil pump is chain driven.

High and low pressure circuits



Fuel is injected into the combustion chamber at high pressure of up to 200 Bar. Before it is injected, the fuel must first pass through two circuits:

- a low pressure circuit, from the fuel tank to the high pressure pump (**A -> B**); fuel enters the pump at a pressure of **5-6 bar**;
- a high pressure circuit, from the pump to the injectors (**B -> C**); pressure varies from **50 to 200 Bar**.